THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Light of speed 1.5×10^8 m/s in a medium has refractive index 1

(a) 2 **(b)** 4.5 **(c)** 1.5 **(d)** 0.5

2. A real image is formed by a convex lens on a screen. If the lower half of the lens is covered by black paper, the image on the screen 1

(a) vanishes **(b)** shows only the lower half

(c) is complete but dimmer **(d)** shows only the upper half

3. Assertion (A): A pencil partly dipped in water appears bent. Reason (R): Light changes its speed on moving from water to air. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. A ray travels from medium A (n = 1.5) to medium B (n = 2). The ray 1

(a) goes straight **(b)** bends away from the normal

(c) is totally reflected **(d)** bends towards the normal

5. A double convex lens of power +4 D is in contact with a double concave lens of power -4 D. The combination 1

(a) has zero power and behaves like a plane glass plate

(b) has -8 D

(c) converges light

(d) has +8 D

6. Assertion (A): The refractive index of a medium is always greater than 1. Reason (R): Light is slowest in vacuum. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. Absolute refractive index of A is 1.5 and that of B is 1.2. The refractive index of A with respect to B is 1

(a) 1.8 **(b)** 0.5 **(c)** 0.8 **(d)** 1.25

8. Two convex lenses of f1 = 20 cm and f2 = 30 cm are in contact. The effective focal length is 1

(a) 12 cm **(b)** 50 cm **(c)** 10 cm **(d)** 25 cm

9. Assertion (A): A convex lens of focal length 10 cm can form a real image of any size. Reason (R): The size of a real image depends on the object distance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A glass slab of thickness t shifts a ray sideways. If the slab is replaced by one of a higher refractive index with same t and angle, the lateral shift 1

(a) decreases **(b)** remains the same **(c)** increases **(d)** becomes zero

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a virtual image never form on a screen but a real one can? 2

12. A convex lens of f = 20 cm produces a real image 3 times the size of the object. Find the object distance. 2

13. Why does the focal length of a lens change with the colour of light? 2

14. An object is moved from infinity towards a convex lens of f = 10 cm. Where does the image form when the object is at 20 cm? Give the nature of the image. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the speed of light is different in different media but its frequency is unchanged. 3

16. A 2 D lens and a -0.5 D lens are placed in contact. Find the focal length of the combination and nature. 3

17. A concave mirror of radius 30 cm is used to form a real image of an object 5 cm tall placed 20 cm from it. Find image size. 3

OR

A glass slab of thickness 6 cm and n = 1.5 is placed over a coin. By how much does the coin appear raised? 3

18. Why can concave mirrors form both real and virtual images but convex mirrors only virtual images? 3

OR

Why are camera lenses not made as a single simple convex lens? Give the reason related to image formation. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A ray enters a rectangular glass slab of n = $\sqrt{2}$ at an angle of 45 degrees. Find the angle of refraction. If the slab is 4 cm thick, discuss qualitatively the lateral shift and why the ray emerges parallel. 5

OR

Explain, with a calculation, why a concave lens of f = -20 cm gives diminished virtual images for objects at 20 cm, 40 cm and 60 cm. 5

20. Using the mirror formula, find the image position when an object is placed 5
(a) at the focus and
(b) midway between pole and focus of a concave mirror. Give the nature and magnification.

OR

Compare a plane, a concave and a convex mirror on the basis of image nature, size, field of view and one use each. 5

END OF QUESTION PAPER Total: 40 Marks